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Best practices for your confirmatory factor analysis: A JASP and *lavaan* tutorial

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Abstract

Confirmatory factor analysis (CFA) is a fundamental method for evaluating the internal structural validity of measurement instruments. In most CFA applications, the measurement model serves as a means to an end rather than an end in itself. To select the appropriate model, prior validity evidence is crucial, and items are typically assessed on an ordinal scale, which has been used in the applied social sciences. However, textbooks on structural equation modeling (SEM) often overlook this specific case, focusing on applications estimable using maximum likelihood (ML) instead. Unfortunately, several popular commercial SEM software packages lack suitable solutions for handling this 'typical CFA', leading to confusion and suboptimal decision-making when conducting CFA in this context. This article conceptually contributes to this ongoing discussion by presenting a set of guidelines for conducting a typical CFA, drawing from recent empirical research. We provide a practical contribution by introducing and developing a tutorial example within the JASP and *lavaan* software platforms. Supplementary materials such as videos, files, and scripts are freely available.

Keywords Confirmatory factor analysis · Structural equation modeling · Internal structure validity · JASP · *Lavaan*

Introduction

In practical research scenarios, evaluating the internal structure of a measurement instrument is a common practice and is often accomplished through the application of factor analysis or item response theory (Bandalos, 2018; Flora & Flake, 2017; Hughes, 2018; Reeves & Marbach-Ad, 2016; Rios & Wells, 2014). Researchers in the social sciences typically operate within a predefined theoretical framework, particularly in empirical applications (Sireci & Sukin, 2013). Confirmatory factor analysis (CFA) has emerged as a pivotal technique in such contexts, offering a comprehensive method for comparing the hypothesized measurement model structure with the observed one (Rios & Wells, 2014). Consequently, CFA enjoys widespread adoption in the field, with approximately 50% of researchers relying on it to evaluate primary data (Crede & Harms, 2019).

CFA is a versatile tool capable of addressing three critical aspects of internal structure: (a) calculation of internal consistency measures, such as composite reliability (McDonald's Omega, for instance); (b) evaluation of model fit across various forms, including congeneric, correlational, higher-order, and bifactor models; and (c) testing of configural, metric, or scalar invariance through multigroup analysis (Rios & Wells, 2014).

This article focuses on aspects (a) and (b) as they align with common practices in applied social sciences (Nye, 2022). Given the depth of these discussions, item (c), which pertains to measurement invariance and other validity aspects (Sireci & Sukin, 2013), falls beyond the scope of this article's objectives.

What exactly do we mean when we talk about a frequent research scenario in applied social sciences? The apparatus used to measure phenomena distinguishes quantitative research in social sciences from fields such as biology, chemistry, earth sciences, and so on (Reeves & Marbach-Ad, 2016). The fact that objective variables are often latent is a crucial element of this instrumentation.

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That is, these variables are not immediately observable and must be deduced from observed behavior (Flake et al., 2017; Reeves & Marbach-Ad, 2016). Furthermore, researchers frequently begin with a pre-existing instrument that is used as a tool to achieve their goals rather than being the goal itself.

Why should a researcher employ CFA when using a measurement model that has been previously validated in prior research, especially when the primary goal is to examine the structural relationships among the studied entities? The concept of validity underscores the collection of 'accumulated evidence', as opposed to relying on a single study that results in a binary 'valid/invalid' verdict. As multiple studies accumulate in support of a particular research hypothesis, the concept gains credibility. The more evidence aligns with the proposed application of a test, the more legitimate its usage becomes (Bandalos, 2018).

It is relevant to acknowledge that the internal structure of a test may vary across different populations in terms of the number of factors or factor patterns (Flake et al., 2017; Flora & Flake, 2017). Therefore, it is advisable to reevaluate a scale's psychometric properties before applying it to a new population, with a specific focus on the factor structure (Flake et al., 2017; Flora & Flake, 2017).

Moreover, when interpreting scores in different contexts, each intended interpretation should be verified. If a user or evaluator provides an interpretation or application of a score that deviates from the test creator's, they are responsible for providing additional evidence of validity (AERA et al., 2014; Flake et al., 2022; McNeish, 2023b).

Psychometric scale scores must furnish diverse validity evidence and sufficient reliability indicators. This evidence must undergo consistent scrutiny to establish the scale's applicability across various contexts, samples, populations, and over time (Valentini & Damásio, 2016). Validation is an ongoing process. As our understanding of a construct evolves or when the instrument is extended to new populations, more validity evidence may be required (Flake et al., 2017; Reeves & Marbach-Ad, 2016).

Considering this continuous validation process, subsequent research should delve into the psychometric properties of existing tests, especially if these tests are introduced to new populations, modified in any manner (e.g., creating a brief version or translating it into another language), or are entirely new. CFA can be a valuable tool in such scenarios (Flora & Flake, 2017).

Another issue of debate is whether a researcher should use exploratory factor analysis (EFA) before or after CFA (Bido et al., 2018). The answer is no, if the researcher possesses a scale with substantial psychometric features that has previously been utilized in similar circumstances and target groups. When the researcher lacks a clear or somewhat comprehensive expectation about the structure of relationships, EFA should be used (Rogers, 2022).

Implementing EFA in situations where CFA would be appropriate risks undermining theories (Bido et al., 2018). When EFA is employed in scientific study, the conceptual meaning of factors is frequently lost. This has serious consequences, such as the lack of result comparability with past research, which is critical for disclosing and discussing the study's contributions in academia, and the disrespect for current theories regarding the issue (Bido et al., 2018).

Bido et al. (2018) identifies certain cases where EFA was applied to known scales when CFA should have been employed, resulting in factors that differed from those established in theory and prior research. The authors found incorrect usage of EFA in 59 percent of the applications studied and advocate expanding the use of confirmatory methodologies to start debates about repeatability and then compare, test, and expand current concepts (Bido et al., 2018).

This prompts the following question: Why should researchers invest their time in employing CFA when their goal is to use structural equation modeling (SEM) to test their research hypotheses and models? The answer lies in the need to thoroughly scrutinize the measurement model before evaluating the structural model within the SEM. This critical step ensures that the measurement variables effectively represent the underlying constructs and factors. When the proposed factors fail to accurately represent genuine constructs, their incorporation into an SEM model lacks merit (Jackson et al., 2009).

In numerous instances, challenges encountered in SEM models stem from deficiencies in the parameters of the measurement model, which CFA can help identify (Brown, 2015). The measurement model is inherently more susceptible to issues than the structural model (Brown, 2023).

One recommended approach for estimating structural relationships in SEM involves a two-step process. In the initial step, the CFA refines the measurement model. Subsequently, the structural relationships were calculated using the complete model, which is a joint model that combines the measurement and structural models (Kline, 2016). This methodology serves to prevent the propagation of measurement errors into the structural model (Jackson et al., 2009). Therefore, even when CFA models are not the primary focus of SEM analysis, establishing a robust measurement model is a prerequisite for gaining insight into the structural relationships among latent variables (Brown, 2023).

Why should someone choose to study CFA from this article when excellent guides are readily available (Brown, 2015; Harrington, 2009; Hoyle, 2023; Kline, 2016)? First, these guides do not provide specific guidance on running CFA models in JASP. Furthermore, the existing JASP manuals (Goss-Sampson, 2022; Navarro et al., 2019; Walker et al., 2022) do not cover SEM or CFA topics. Although several comprehensive *lavaan* manuals are available (Beaujean, 2014; Finch & French, 2015; Gana & Broc, 2019; Rosseel, 2023), they do not offer the best practices for a typical CFA.

Second, statistical guides often lag behind the most recommended practices because of the rapid production of scientific information. Consider the remarkable Dynamic Fit Index (DFI) proposal, the core study of which was published in the same year as this article (McNeish, 2023b; McNeish & Wolf, 2023; Wolf & McNeish, 2023), with ongoing developments (McNeish, 2023a; McNeish & Manapat, 2023). Furthermore, most SEM guides continue to allocate more space to models constructed from continuous items and estimated using maximum likelihood (ML). However, many practical social science researchers work with ordinal observed measures, for which ML is not recommended (Bandalos, 2014; Beauducel & Herzberg, 2006; DiStefano & Morgan, 2014; Flora & Curran, 2004; Forero et al., 2009; Holgado-Tello et al., 2016; Holgado-Tello et al., 2018; Kiliç et al., 2020; Kiliç & Dogan, 2021; Li, 2016a, b; Mîndrilă, 2010; Morata-Ramírez & Holgado-Tello, 2013; Nye, 2022; Savalei & Rhemtulla, 2013; Yang-Wallentin et al., 2010). In addition, several of these guides are directed at software that still lacks substantial functionality for working with ordinal variables.

In this context, manuals often serve as the primary gateway for novice users. This may contribute to the prevalence of suboptimal reporting and application of CFA methods in reputable journals, emphasizing the need for articles that provide up-to-date recommendations (Crede & Harms, 2019; Jackson et al., 2009) and best practices (Nye, 2022; Schmitt, 2011).

Based on these arguments, we encourage readers to continue exploring the content presented in this article. Subsequently, our focus will shift to a concise discussion of recommended techniques for conducting a typical CFA in the context of applied social sciences, drawing from the latest empirical research (Crede & Harms, 2019; Nye, 2022). In Section "The Software", we discuss the considerations related to the software employed for these analyses. Building on the concepts introduced in Section "Best practices for typical CFA", the results section offers an illustrative tutorial on conducting a typical CFA within the realm of applied social sciences, complete with visual representations on the JASP screen. Additionally, we provide a supplementary *lavaan* tutorial that not only replicates but also enhances the JASP results, presented comprehensively through videos and R scripts. The JASP tutorial is also accessible as supplementary material, featuring videos and the .jasp file. In conclusion, the final section provides closing remarks and summarizes the key insights offered in this article.

Best practices for typical CFA

CFA plays a pivotal role within the framework of covariance-based structural equation modeling (CB-SEM) (Brown, 2023; Harrington, 2009; Jackson et al., 2009; Kline, 2016; Nye, 2022). To provide meaningful recommendations for CFA, we must place them within the context of CB-SEM. While the best practices for CB-SEM are generally applicable to CFA, this article will specifically focus on elucidating them in the context of measurement models in CB-SEM. It excludes adjustments commonly found in variance-based SEM (VB-SEM) resources, which are more relevant for fine-tuning measurement models in the partial least squares SEM (PLS-SEM) literature (Hair et al., 2017, 2022; ; Henseler, 2021).

Please note that this article does not provide an exhaustive exposition of CB-SEM or CFA. For a comprehensive understanding of CB-SEM, it is recommended to consult trustworthy guides (Brown, 2015; Harrington, 2009; Kline, 2016; Whittaker & Schumacker, 2022) and scientific publications that offer robust introductions to CB-SEM (Davvetas et al., 2020; Shek & Yu, 2014). These resources typically cover CB-SEM using conventional procedures of specification, identification, estimation, and evaluation. Assuming that readers have a fundamental understanding of the subject and are guided by the definition of a typical CFA in applied social sciences, this article addresses the following aspects: i) best practices (Flake et al., 2017; Nye, 2022); ii) what to avoid (Crede & Harms, 2019); and iii) what to include in your reporting (Flake & Fried, 2020; Jackson et al., 2009).

Due to the pragmatic nature of this article's recommendations, which aim to offer clear, concise, and objective guidance for conducting a typical CFA in applied social sciences, we have chosen to present the recommended best practices, what to avoid, and reporting guidelines in the form of an Appendix Table 3. This approach instructs researchers on how to minimize the risk of chance capitalization in a typical CFA within the field of applied social sciences.

The Software

JASP (JASP Team, 2023) is versatile software compatible with Windows, MacOS, and Linux. It is often described as an R front-end (R Core Team, 2022). This open-source tool has gained recognition as a valuable and free alternative to proprietary software widely used in applied social sciences (Navarro et al., 2019) and is recognized as one of the leading open-source options (Love et al., 2019). Notably, it has made its mark in undergraduate education, proving beneficial for teaching quantitative analysis with empirical data (Agawin, 2020; Campayo et al., 2022).

JASP offers several notable features. The user interface is innovative, featuring drag-and-drop functionality, accessible menus, and minimalistic outputs. It computes results in real time as users adjust options, streamlining the analytical process. In addition, it facilitates peer review without the need for complex syntax (Goss-Sampson, 2022).

One of JASP's significant distinctions, particularly in the realm of empirical applications in social sciences, is its robust support for Bayesian analysis (Niu et al., 2021; Wagenmakers et al., 2018). A quick search on Google Scholar using the "allintitle: JASP" filter reveals that most scientific papers employing JASP focus on Bayesian inferences.

The modular nature of JASP allows users to install specific modules based on their needs. It covers various statistical techniques, including descriptive, *t* tests, ANOVA, mixed models, regression, frequencies, and more. In JASP, CFA functionality is housed within the *Factor* module (*Factor/CFA*), which serves as a front-end for *lavaan* (Rosseel, 2012), one of the most commonly used R packages for SEM. However, with the release of version 0.17.1, it is now possible to fit a CFA model directly within JASP's SEM (*SEM/SEM*) module using the *lavaan* syntax (Fig. 1b).

As of version 0.17.1, JASP has introduced two notable features, each with its own advantages and limitations. On the negative side, JASP does not support data modification, necessitating the use of other software for this purpose. For example, double-clicking on the database screen opens Excel for data editing, and users can choose their preferred spreadsheet editor in the *Data preferences* (Fig. 1c). JASP supports a limited number of file formats (*.jasp *.csv *.txt *.tsv *.sav *.ods *.dta *.por *.sas7bdat *.sas7bcat *.xpt), making it advisable to prepare data with other applications and generate raw databases in formats such as *.csv or *.txt for JASP analysis. When changes are made to the original data file, they can be synchronized with the database, or automatic synchronization can be enabled in the *Data preferences* menu (Fig. 1d).

A valuable feature of JASP is that all analysis results are stored in a single database file, simplifying data management, especially for those new to data analysis. Users can customize titles, add notes, extract citations from the R packages used, copy analyses for editing in tools such as Excel, Google Sheets, or Latex, and replicate analyses from the output screen. The function for replicating results in APA format with customizable layouts is highly valuable (Chinchu, 2022), as is the capability to minimize repetition when conducting similar analyses with slight variations, thereby facilitating collaborative work (Walker et al., 2022).

Why select JASP vs *lavaan* when they both get the same result? Because of its point-and-click method and drop-down menus, it is simpler for inexperienced users (Goss-Sampson, 2022; Navarro et al., 2019; Walker et al., 2022) or researchers in applied social sciences who typically lack a broad skill set. These individuals usually experience difficulties with quantitative methodologies or programming scripts (Fiates et al., 2014; Flake et al., 2017).

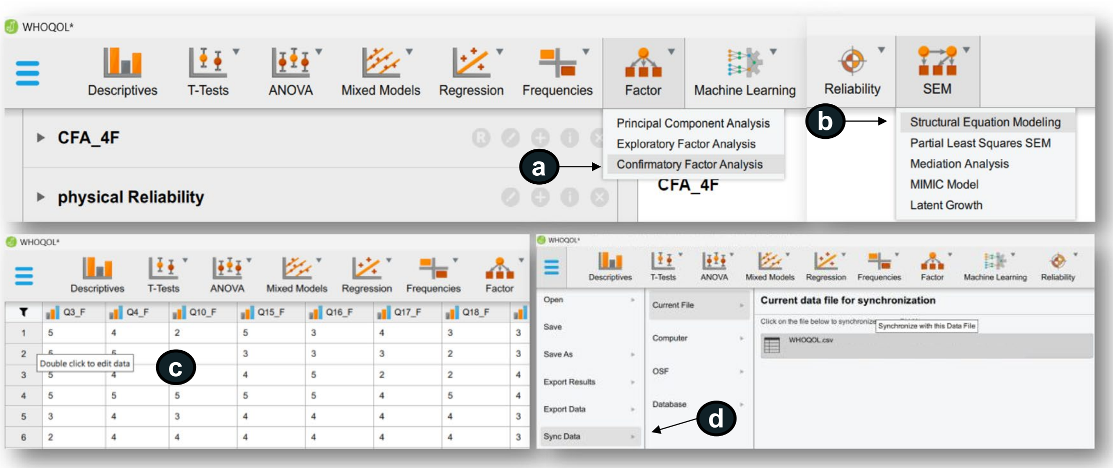

Fig. 1 Key modules for CFA, Visualization, and data synchronization in JASP. **Note:** (**a**) *Factor/CFA* menu; (**b**) *SEM/SEM* menu; (**c**) Data visualization; (**d**) Data synchronization from the original database

Why should one use *lavaan* if JASP is more user-friendly? JASP is a developing program that is limited to the options accessible in its module, which may not always fully utilize the capability of the underlying R package. As a result, while doing a CFA, individuals with some command-line experience will find that utilizing *lavaan* within R provides them with more flexibility in their analysis (Rosseel, 2023).

Moving forward, we will focus on presenting a typical CFA using sample data from JASP version 0.17.1. Supplementary materials will include the same and additional analyses for *lavaan*. Users can access the *lavaan* commands (along with any other R package used in the JASP module) at any point during the analysis process by clicking on the *Show R syntax* selection (Fig. 2a). In some modules, such as the *Factor/CFA* module, users can extract the syntax of the underlying R package (Fig. 2b).

Tutorial on JASP

In this section, the JASP tutorial is structured into several parts in accordance with the guidelines outlined in the Appendix. The initial part, encompassing steps 1 and 2, will provide an in-depth understanding of the measurement models and elucidate the data pre-processing procedures that were previously conducted in the original study, from which the database was derived. Subsequent sections will discuss the quantitative analyses that extend across the four distinct phases. However, owing to certain limitations within JASP, the final two procedures detailed in the Appendix, specifically 'Model Comparisons and Modifications' and 'Power Analysis,' are primarily performed using *lavaan*.

Throughout this tutorial, the primary emphasis will be on elucidating the key features within JASP and the pivotal decisions required to successfully conduct a typical CFA. The outcomes of these analyses will be subject to further examination in the supplementary material, which includes informative videos, an annotated R script, and a JASP file.

Selection of the measurement model and pre-processing

The database used in this study is the same as that employed by Rogers (2022) in his research on EFA. This choice was made for three primary reasons: (1) The database is freely accessible (Rogers, 2021a, 2021b); (2) It serves didactic purposes and facilitates comparability, enabling readers to refer to Rogers (2022) and compare the results with those presented in this article; and (3) It aligns with current guidelines for analyzing latent variables using the two primary statistical methods, EFA and CFA, while benefiting from a scale with robust psychometric properties in a large sample ($n = 1047$).

The scale used in this study is the World Health Organization Quality of Life-Abbreviated Version (WHOQOL-Bref) by the WHO, which is presently one of the most widely adopted models for assessing quality of life. It has been implemented in more than 50 countries across diverse populations (Lin & Yao, 2022; Skevington & Epton, 2018).

The WHOQOL-Bref comprises 26 five-point Likert items, although only 24 of them were used in this study.

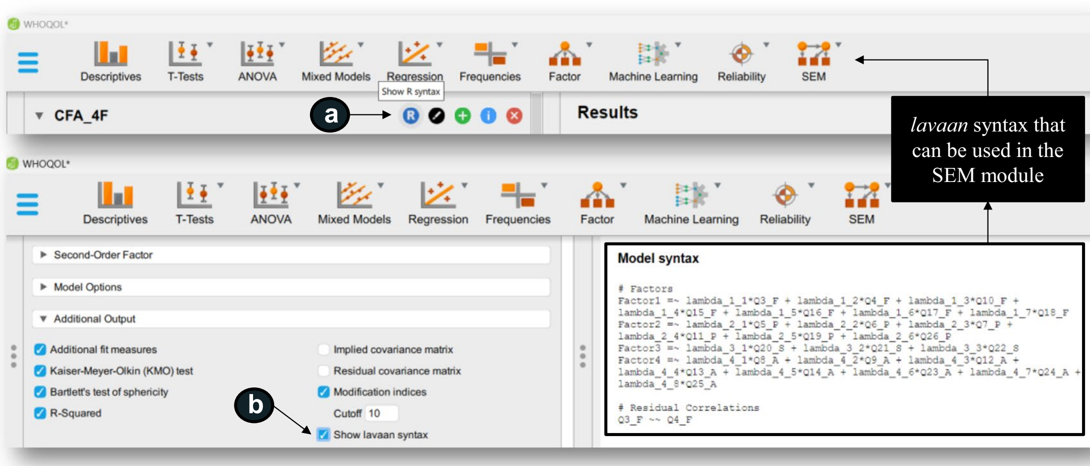

Fig. 2 Syntax for replicating results in *lavaan* or the *SEM/SEM* module. **Note:** (**a**) R syntax available in all JASP menus; (**b**) *lavaan* model syntax available in the *Factor/CFA* module

The scale was designed to assess four domains: physical, psychological, social, and environmental. Extensive systematic reviews and meta-analyses have consistently supported this structure (Lin & Yao, 2022; Skevington & Epton, 2018), affirming its reliability as an instrument with minimal variability across studies (Lin & Yao, 2022). Importantly, it can detect meaningful and substantial variations in quality of life assessments among diverse cultures that employ it (Skevington & Epton, 2018).

Although the four-correlated factor multidimensional structure is the most widely accepted for the WHOQOL-Bref, recent studies have introduced, tested, and substantiated alternative structures, including a five-factor model (Lin & Yao, 2022), as well as more intricate ones such as bifactor or higher-order structures (Perera et al., 2018). In certain populations and cultures, such as a study by Perera et al. (2018) involving a sample of 539 individuals from a general population in an Australian city, other factor structures, including one-dimensional, six, seven, or eight factors, have been identified. However, for this study, we have chosen to rely on the findings of systematic reviews and meta-analyses, which encompassed more than 100 papers from various countries (Lin & Yao, 2022; Skevington & Epton, 2018).

The WHOQOL-Bref has been widely employed across diverse population groups in Brazil (Kluthcovsky & Kluthcovsky, 2009), where the database was collected. As of February 24, 2023, a quick search on Google Scholar, filtered by the term 'allintitle: WHOQOL Bref', indicated that publications in Portuguese alone have yielded over 200 applications of the WHOQOL-Bref in various settings. We initially selected the WHOQOL-Bref because it aligns with the data from the meta-analysis by Lin & Yao (2022) and then considered additional models such

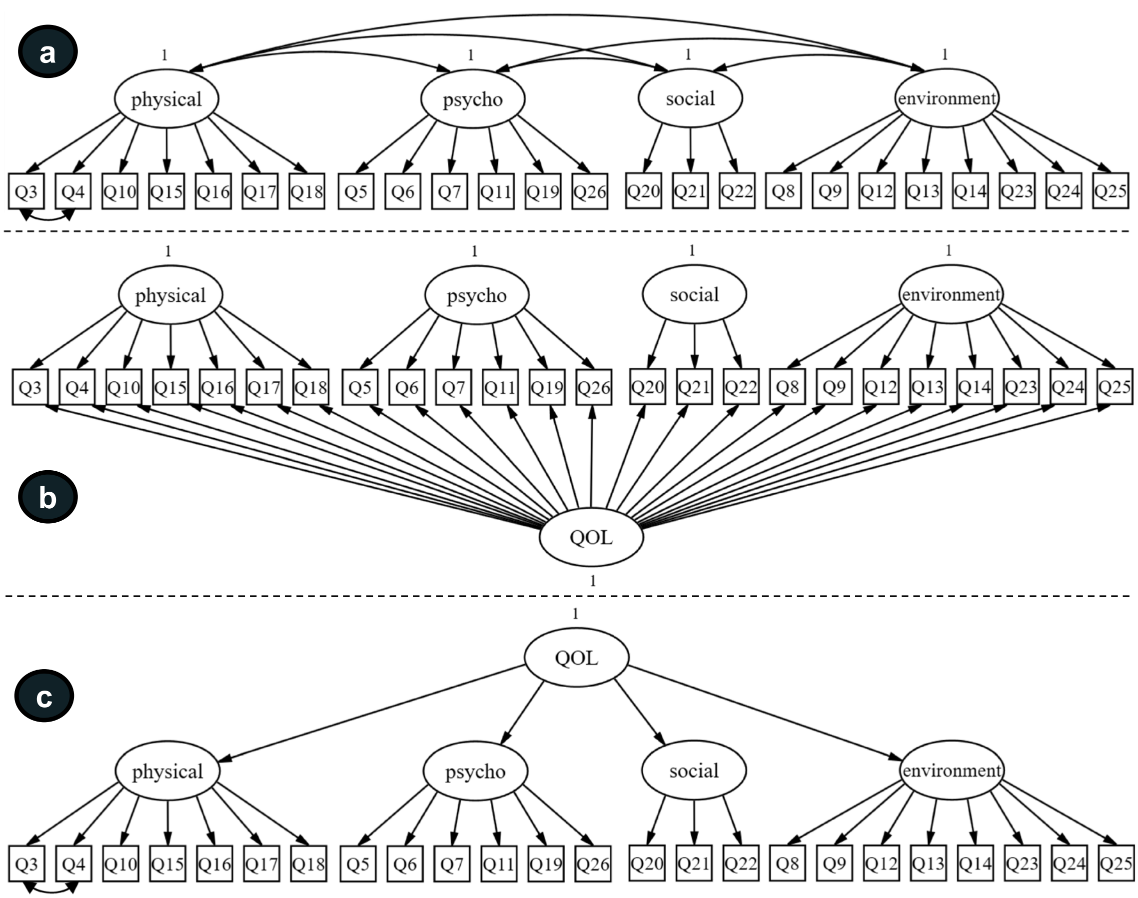

Fig. 3 WHOQOL-Bref factor models under consideration. **Note:** To improve visibility, errors have been deleted from the figure; the value 1 above the latent variable shows variance fixation = 1; **a** four-correlated-factor model predicting the correlation between errors in items Q3 and Q4, as they are two reverse items of the same factor; **b** four-factor model with one general factor (bifactor model); **c** four-factor second-order model (2nd-order model) predicting the correlation between errors in items Q3 and Q4, as they are two reverse items of the same factor

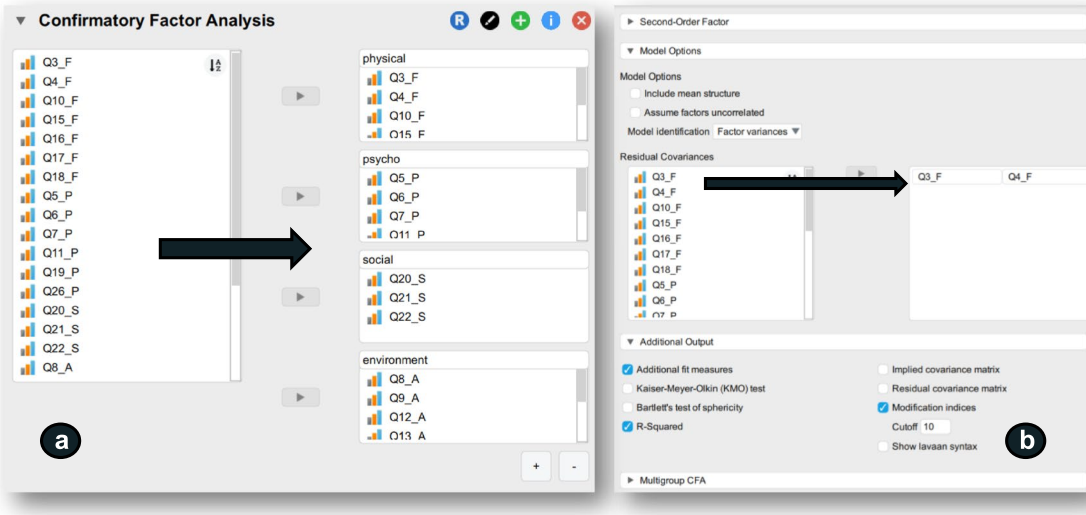

Fig. 4 JASP definition of the 4F-model and output selection. **Note:** **a** The *Factor/CFA* module's main menu, where the components of each factor were set; **b** the *Model Options* and *Additional Output* submenus, for predicting correlations between items, the latent variable scale criterion, and some additional fit measures.

as the bifactor and second-order models, in accordance with the perspective presented by Perera et al. (2018). The three CFA models discussed in this tutorial are visualized in Fig. 3.

Rogers (2022) conducted extensive pre-processing on the initial database, addressing missing data, discrepancies, and engagement issues. Initially, 1558 observations were available, which were subsequently reduced to 1169 after rectifying discrepancies and implementing quality control filters to check for attentive responses. To deal with missing values, a combination of pairwise deletion, listwise deletion, and multiple imputation techniques was used. Multiple imputation was performed using an algorithm that leverages the Markov Chain Monte Carlo (MCMC) method for Bayesian inferences and numerical integration. Consequently, the final sample consisted of 1047 observations, and no signs of multicollinearity were evident among the WHOQOL-Bref scale indicators.

Estimation process

Proceeding with the four-factor correlation model, denoted as 4F-model (Fig. 3a), as illustrated in Fig. 4a of the *Factor/CFA* module, the final step involves the allocation of items to their expected factors (Fig. 1a). The variables in this illustration can be labeled optionally (physical, psychological, social, and environmental) and added by clicking the plus (+) sign in the lower right corner. Depending on the model's complexity, it may be more efficient to make selections first and then define the model because JASP supports real-time computation. The *Second-Order Factor*, which is a submenu beneath the model specification, is not discussed in this section.

In this model (Fig. 3a), we estimate the association between the WHOQOL-Bref items Q3 and Q4. This is achieved in the *Model Options* submenu by selecting two items (Ctrl + right mouse button) and dragging them to the box on the right (Fig. 4b). Additional results are available alongside the software defaults at the bottom right corner of Fig. 4b, under the *Additional Output* submenu. It is worth noting that the Kaiser–Meyer–Olkin (KMO) test and Bartlett's test of sphericity has been increasingly disregarded when analyzing ordinal data because they relate to Pearson correlation matrices and may have limitations.

Certain indices from the *Additional Selection Fit Measures* submenu will only be accessible when using an ML estimate technique because they pertain to information criteria derived from the discrepancy function. To obtain information on significant global fit indices, this option must be chosen. In this submenu, we also select the *R-squared* and *Modification indices*, with the Cutoff set to ten for clearer results. For smaller samples, a cutoff of 3.84 may occasionally be used to indicate when the χ^2 statistic becomes significant, which may warrant the inclusion of that path in the model. The *Multigroup CFA* submenu was not used in the analysis.

The *Misfit plot* from the plot submenu (Fig. 5a) was specifically chosen. This option displays the residual

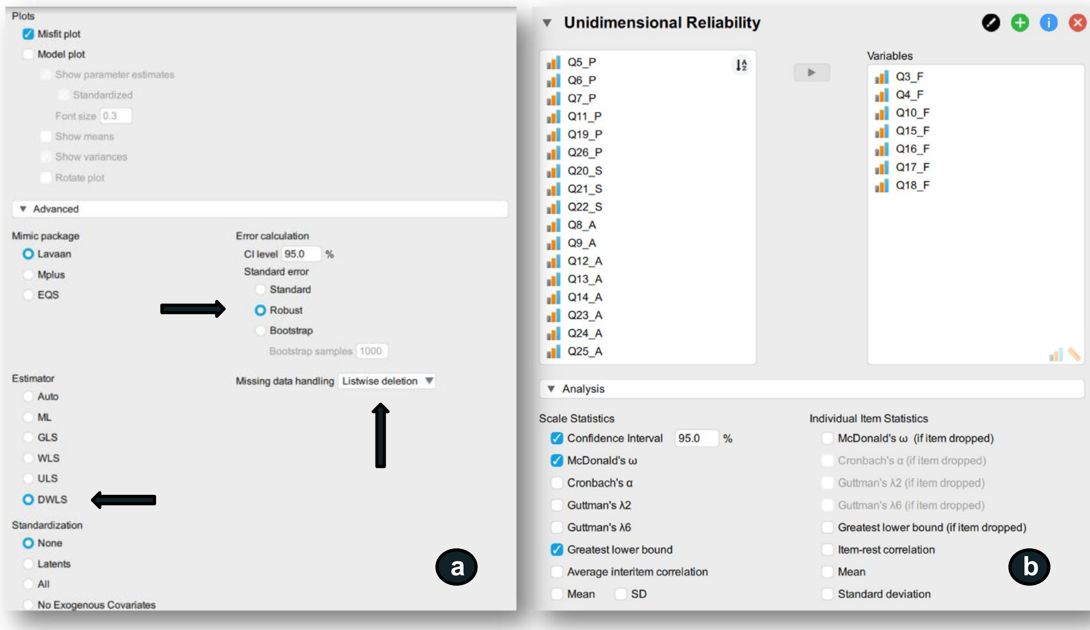

Fig. 5 Physical factor estimator selection and reliability coefficients. **Note: a** *Plots* submenu and *Advanced*, for viewing the residual correlation matrix and selecting estimate techniques; **b** JASP's *Reliability/Unidimensional Reliability* module main menu and *Analysis* submenu, for selecting reliability coefficients.

correlations between elements in a conditionally structured matrix, with darker shades indicating stronger correlations (ideally less than 0.10). The *Model Plot* option is less practical for complex models with many items because it does not produce visually appealing graphics. In most SEM tools, diagrams are primarily used for examination purposes, and researchers are generally required to design their models for presentation in articles or reports using graphic editing software.

A crucial decision in a typical CFA is selecting the *DWLS* estimator and *Robust* standard errors from the *Advanced option* (Fig. 5a). This choice is based on a reasonably large sample size. However, the *ULS* estimator may be suitable for smaller samples. In cases where the FIML option is set to *Missing data handling*, an error warning will appear because this missing data treatment is only available when using an ML estimator with continuous data. In this context, the choice is to remove all cases containing any missing observations (*Listwise deletion*), especially because the missing data in this database were resolved during pre-processing.

No specific *Standardization* is required because all scale items share the same unit of measurement, and the variance of the latent variables has already been set to 1. All options can be considered in this regard. for the *DWLS* estimator to work effectively with ordinal data, it must first be defined as ordinal in data visualization (Fig. 1c). If the variables are defined as scalars, the output values will mirror those from the *cfa* parameter in *lavaan* (*ordered = FALSE*), indicating that JASP will employ the *DWLS* estimator for continuous data.

The steps described above complete the 4F-model estimating process. It should be noted that there were no problems with algorithm convergence or model identification. To save space, standardized and non-standardized parameters are not mentioned in the article but may be found in the supplemental materials. Furthermore, in the *Reliability/Unidimensional Reliability* module, we must compute McDonald's

Table 1 Dynamic Fit Index (DFI) for the population model

Level	SRMR	RMSEA	CFI
0	0.025	0.012	0.998
1	0.037	0.036	0.986
2	0.064	0.084	0.933
3	0.079	0.105	0.904

Specificity is 95%. Values generated using the *dynamic R package* using the *catHD* function, with models estimated by WLSMV/DWLS (McNeish, 2023a) and $n = 1047$

Omega and GLB for each of the four factors (Fig. 5b). In the *Analysis* submenu, we choose these reliability coefficients.

Some of the *Individual Item Statistics* choices may be considered. Another option to examine later in the *Advanced* submenu is to estimate the confidence interval using bootstrapping. Consider this option for your results and for publishing them. Enabling all bootstrapping options available in SEM analyses with real-time computing might cause PC RAM to be overloaded and file storage size to grow. Even if some menus allow you to choose not to store bootstrap samples, there may be difficulties with the file and probable crashes.

4F-model fitting

After scrutinizing variances and standardized loadings, no Heywood instances were detected, aligning with the absence of convergence and identification issues. The conventional threshold values from JASP, which encompass SRMR = 0.057, RMSEA = 0.065, and CFI = 0.985, signify a favorable overall model fit. Nevertheless, it is prudent not to rely solely on these traditional thresholds, particularly since JASP version 0.17.1 does not offer scaled measures, leaving us with only the SRMR as a dependable indicator.

For a more comprehensive evaluation of model fit, we turn to the dynamic cutoffs (DFI) outlined in Table 1 (McNeish, 2023a; McNeish & Wolf, 2023), as expounded in the supplementary materials. These dynamic cutoffs are grounded in a population model proposition deduced from the meta-analysis conducted by Lin & Yao (2022) and using the *R dynamic package*. According to these dynamic benchmarks, the 4F-model registers a level 2 (mediocre) misfit based on the SRMR value of 0.057. Although this may be initially acceptable, the individual factors, Physical (0.819, 95% CI 0.803–0.836), Psychological (0.823, 95% CI 0.806–0.839), Social (0.696, 95% CI 0.665–0.727), and Environmental (0.789, 95% CI 0.769–0.808), exhibit commendable Omega coefficients. Furthermore, the GLB coefficients surpass the Omega coefficients, indicating sound reliability.

The model has a modest factor loading of $\lambda_{Q4} = 0.423$ and another acceptable factor loading of $\lambda_{Q24} = 0.482$. There were a few strong residual correlations (> 0.10) that were connected to item Q5, which was also the origin of three extremely high (>100) modification indices (and EPC), showing cross-loading with Q14 and high loading on Environmental and Physical factors. The projected connection between Q3 and Q4 (= 0.388) was

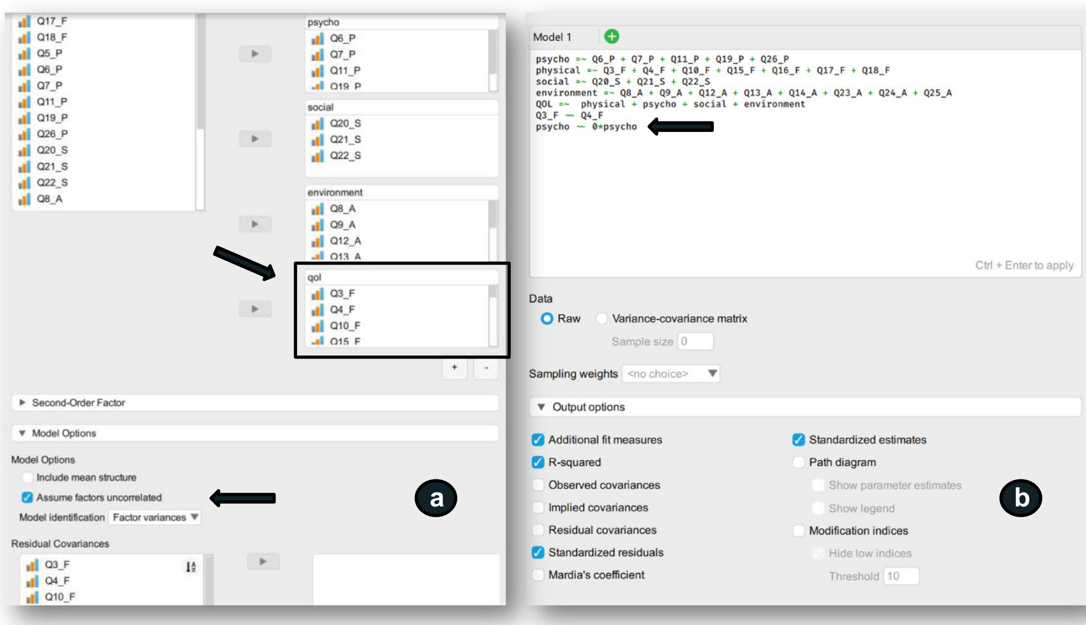

Fig. 6 Bifactor and 2nd-order model. **Note: a** The *Factor/CFA* module's main menu, with the general factor prediction and *Model Options* submenu to define the hypothesis of uncorrelated factors for model identification; **b** Second-order model definition in the *SEM/SEM* main module, with predicted variance = 0 for the latent variable Psycho for identification purposes, and outputs selected in the *Output options* submenu

extremely significant and meaningful. Despite being subject to improvement and the warning directed at item Q5 ("How much do you enjoy life"), we can initially adhere to the 4F-model given an acceptable effect size (level 2) and strong evidence both nationally and internationally in different contexts favorable to the structure of four correlated factors.

The key warning, however, is with regard to the global fit indices supplied by JASP: as of version 0.17.1, JASP only gives traditional values (not scaled or robust). Despite the fact that *Robust* is selected in the *Advanced* option, this technique only applies to the parameters' standard errors. This will only be handled when we employ *lavaan*, which will be detailed further below.

Model comparisons and modifications

To estimate the bifactor model (Fig. 3b), a similar approach to the previously described estimation process is followed. However, instead of predicting a general factor that encompasses all items, it is assumed that the factors are uncorrelated to identify the model (Fig. 6a). The other options remain unchanged. For identification purposes, item Q5 needed to be omitted in this example and in the second-order model, as it posed challenges in the previous analysis. Omitting this item was done to maintain consistency and facilitate comparison.

The second-order model (Fig. 3c) can be estimated by predicting the second-order factor in the *Second-Order Factor* submenu. However, the second-order model was constructed and evaluated in the *SEM/SEM* module (Fig. 6b) because the variance of the latent variable Psycho had to be fixed at 0 for identification purposes. Building models in the *SEM/SEM* module offers greater control over the parameters to be calculated, as they are created using *lavaan* syntax (Fig. 1b). In this module, *Additional fit measures*, *R-Squared*, *Standardized residuals*, and *Standardized estimates* are selected from the *Output options* submenu. If further model investigation is required, *Modification indices* may also be utilized (Fig. 6b).

Multiple models for comparison can be included in the *SEM/SEM* module. As demonstrated in Fig. 7, we incorporated the 4F-model, along with the other two multidimensional models, in this comparison, all without item Q5. Figure 7a displays the syntax of the 4F-model, Fig. 6b shows the syntax of the second-order model, and Fig. 7b presents the syntax of the bifactor model. Additionally, as seen in Fig. 7, names can be assigned to the models to enhance visibility and output formatting.

Once the *lavaan* syntax is input, and the estimators are selected in the *Estimation options* submenu, the results can be quickly reviewed by pressing Ctrl + Enter on the keyboard. For refining the model and obtaining results with standard errors based on *Bootstrap*, the *Estimation options* submenu can also be accessed. There are no options to edit in the *Model Options* and *Multigroup SEM* submenus. The settings chosen in these submenus will apply to all

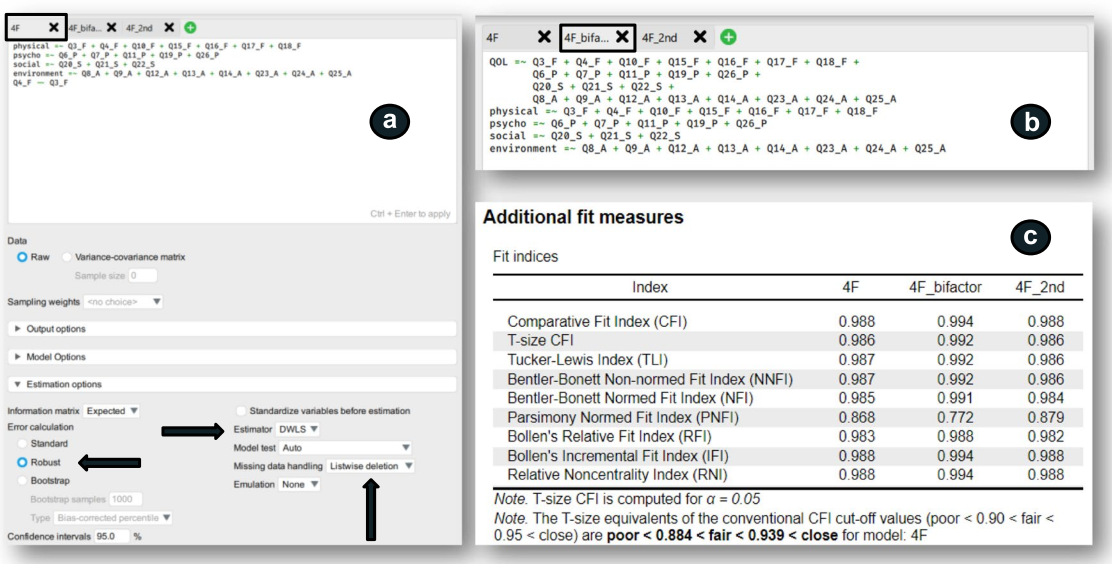

Fig. 7 Model comparison using *SEM/SEM Modeling*. **Note: a** 4F-model syntax without item Q5 and estimation technique selection for all models; **b** bifactor model syntax without item Q5; and **c** example of one of the outputs for comparing the models

Table 2 Global fit of models based on the scaled χ^2 statistic

Index	4F-model A	4F-model B	Bifactor-model	2^{nd}order-model
χ^2	1,732.820	1,285.457	1,212.683	1,285.170
p value	0.000	0.000	0.000	0.000
Df	245	223	207	226
χ^2/df	7.073	5.688	5.858	5.687
CFI	0.941	0.956	0.958	0.956
TLI	0.934	0.950	0.949	0.951
SRMR	0.057	0.053	0.049	0.054
RMSEA	0.076	0.067	0.068	0.067
[95% CI]	[0.073–0.080]	[0.064–0.071]	[0.064–0.072]	[0.063–0.071]

Values discovered using *lavaan* script analysis, which is presented in the supplemental material. 4F-model A is the original WHOQOL scale (Fig. 3a), while 4F-model B represents the model in Fig. 3a without item Q5

three models, as individual settings for each model are not available. It is important to note that, at least until version 0.17.1, JASP does not provide assessments of reliability for multidimensional models such as the bifactor and second order. Instead, unidimensional Omega values can be computed in the *Reliability* module, and these calculations may differ for general and second-order factors in a CFA context. Detailed values can be found in the supplemental material.

The encouraging aspect is that evidence of a general "quality of life" factor exists, and a general scale score can be generated. This is a common practice in several applications, but it often does not validate the presence of a general or second-order factor. In this approach, all items cannot be simply summed or averaged. Instead, the general score must be estimated within the framework of a bifactor or second-order model, as demonstrated in this CFA example.

However, it is essential to treat the results of these comparisons with caution. JASP version 0.17.1 does not provide scaled global fit indices, regardless of the choices made in the *SEM/SEM* submenu (*Estimation Options/Error Calculation* and *Model Test*) or the *Factor/CFA* submenu (*Advanced/Standard Error*). This limitation should be considered when interpreting the results.

Lavaan and power analysis

In this section, the same analyses as previously described are conducted using *lavaan*, along with an annotated R script available in the supplemental material. This approach offers additional analyses and measurements that are not accessible in JASP version 0.17.1. These additional capabilities include assessing the reliability of higher-order and bifactor models, obtaining scaled global fit indices, generating visually presentable model graphics, calculating factor scores (particularly vital for bifactor or second-order models), and using the R ecosystem through multiple packages. It is worth noting that while JASP lacks power analysis for SEM models, such an analysis can be executed using the *simsem* package within the R script (Pornprasertmanit et al., 2022).

The results derived from the analysis in JASP are replicated in the lavaan-based script. It is essential to consider the *Robust* column rather than the *Standard* column for evaluating scaled global fit values, as delineated in Table 2 for educational purposes. This allows for a comparison of effect sizes, specifically regarding the degree of fit/misfit, based on RMSEA and CFI, similar to the previous assessment using SRMR. These three indices collectively indicate a comparable degree of fit/misfit, with all models falling within Level 2 (mediocre) from an effect size perspective. This suggests that the examined models appear to be on par, even though the original model (Fig. 3a) exhibits slightly poorer fit.

The example presented in this tutorial underscores the critical role of an exhaustive literature review when selecting a measurement model. In terms of the degree of fit/misfit concerning the population model, the models under consideration exhibit statistical closeness, as determined through power analysis. However, there is a lower preference for the original 4F model (Fig. 3a).

From a practical perspective, employing a bifactor model (Fig. 3b) can be advantageous when the primary aim is to justify the use of a single "quality of life" factor without intricate scrutiny, and the goal is solely to compare factor scores. Nevertheless, when the intention is to incorporate the measurement model into a structural model, the complexity of the task increases. The relatively better performance of the bifactor model, as assessed through the conventional method of comparing models using point values of global fit indices, aligns with existing literature. This suggests that while bifactor models involve a greater number of parameters, they frequently offer better fits. Some researchers even argue that bifactor models have been employed to salvage traditional scales that have become obsolete due to new estimation and factor selection methodologies. It is important to note that bifactor models assume that certain factors are uncorrelated, a condition that may not always hold true in reality.

Therefore, based on compelling empirical evidence favoring the 4F-model (Lin & Yao, 2022), the potential for bifactor models to exhibit tendencies towards overfitting, and a power analysis approach that indicates similar degrees of fit/misfit, our preference would be to select a model with four correlated factors (Fig. 3a), excluding item Q5, for further research.

As the data has already been collected, it is advisable to conduct a post hoc power analysis. While an a priori power analysis would be ideal for sample size calculation (Feng & Hancock, 2023), it would need to account for the structural model, which falls beyond the scope of this article. In JASP, it is not feasible to calculate sample power from an SEM model using simulation or analytical solutions. Therefore, we recommend utilizing the *simsem R package* (Pornprasertmanit et al., 2022), as outlined in the supplemental material, to conduct a post hoc power analysis of the original 4F-model (Fig. 3a), excluding item Q5. This analysis takes into consideration the ordinal characteristics of the items and the sample size (n = 1,047). The script provides comprehensive information on estimated parameter bias, standard error bias, coverage, and power. All data points fall within the acceptable ranges for the parameters of interest, as detailed in the Appendix.

Concluding remarks

This tutorial underscores the paramount importance of adhering to best practices in CFA while navigating the constraints of available software tools. We meticulously followed the fundamental guidelines outlined in the Appendix, aligning them with the example at hand. Even though the recommendations of Step 2 had already been implemented in the initial study that generated the dataset for this analysis, we remained committed to applying these guidelines, notwithstanding the limitations of JASP in terms of CFA model manipulation, Dynamic Fit Index (DFI) calculations, and power analysis. We highly recommend incorporating the *lavaan* R package for research that ventures beyond the capabilities of JASP.

In our pursuit of scientific rigor, certain recommendations, such as the presentation of standardized and nonstandardized coefficients, were omitted from the article to save space, but they can be found in the supplemental materials. This approach also serves to underscore the limitations of relying solely on a single software program to cater to a researcher's diverse needs. While proprietary software such as Mplus offers comprehensive tools for typical CFA, it can be financially prohibitive for some academics. On the other hand, other proprietary solutions, such as AMOS, lack robust features for handling ordinal data, and software like LISREL and EQS can be less user-friendly.

Lavaan, as part of the R ecosystem, possesses a steeper learning curve but stands as a free and feature-rich alternative to Mplus. JASP, functioning as a user-friendly frontend for R, streamlines the learning curve for employing *lavaan*. However, it currently falls short in providing all the necessary capabilities for implementing best practices in a typical CFA. The absence of scaled global fit measures, multidimensional composite reliability measures (Omega), DFI calculations, and power analysis remains a significant limitation. The limitation of not including the DFI is not as harsh because it is a new proposal that must still be carefully considered.

Nonetheless, research suggests that some of these limitations can be surmounted by treating scales with five or more response items as continuous variables, utilizing the ML estimator and its robust variations in JASP (Flora, 2020; Lei & Shiverdecker, 2020; Maydeu-Olivares, 2017; Rhemtulla et al., 2012; Robitzsch, 2022). With this approach, we can incorporate FIML for missing data treatment within the CFA and apply Equivalence Testing measures (EQT) (Marcoulides & Yuan, 2017; Yuan et al., 2016). T-size CFI and T-size RMSEA can serve as alternatives to DFI, potentially replacing traditional cutoffs.

The evaluation of global model fit in models estimated by the OLS family is more difficult than in models calculated using ML (Robitzsch, 2022). Furthermore, the latter models are older, making them more recognizable to researchers (Rhemtulla et al., 2012). Although models estimated by the OLS family are fast gaining popularity, they remain newer, less known, and to date, less available in software (Rhemtulla et al., 2012).

To further enhance JASP's potential, we have included a JASP file that takes continuous elements into account in the supplemental content, estimating models using robust ML. By employing T-size CFI and T-size RMSEA, we find that all models exhibit reasonable fits, except for the original 4F-model (with item Q5), which is considered a poor fit. Sample size requirements can be assessed using Hoelter's Critical N (CN) for post-hoc power analysis, available in both the *Factor/CFA* and *SEM/SEM* modules.

Addressing the final challenge, global fit power analysis is feasible but not yet available in JASP. Options include using the non-central chi-square test and RMSEA statistics of well-specified or misspecified models. Tools such as the *ShinyApp* by Jak et al. (2021) and the *WebPower* application from Zhang & Yuan (2018) offer solutions based on Monte Carlo simulation. These tools facilitate considerations for missing values and non-normal data and provide a more visual, diagram-based approach. In the R environment, functions from the *semTools* (Jak et al., 2021) and *semPower* packages (Jobst et al., 2023) can be utilized.

In essence, the core focus of this article is not on the simultaneous use of multiple software packages but on emphasizing recent best practices, what to avoid, and what to report in the context of a typical CFA, as outlined in the Appendix.

Appendix

Appendix: Best practices in a typical CFA in applied social sciences under appendix section

Table 3 Best practices in a typical CFA in applied social sciences

Step	Best practices	What not to do	What to report
(1) Measurement model selection	Prioritize scales that have been utilized in the research's target population and have already been verified in the language of application (Flake et al., 2017). Make certain that these measures are neither proprietary or exclusive to a certain specialty, such as psychologists.	Avoid using measuring models that are intended for a certain purpose but lack psychometric rigor (Flake & Fried, 2020; Kline, 2016). Be cautious, since many scales have this feature. The mere existence of a scale does not make it legitimate (Flake et al., 2017).	Justify the strong psychometric qualities of the chosen scale by presenting past empirical evidence of its use in the target population as well as measurements of reliability and validity (Flake & Fried, 2020; Jackson et al., 2009; Kline, 2016).
	Conduct an initial search for the Psychological Test Assessment System in your country. If you cannot find suitable scales, consider those funded by universities, research groups/centers, public or non-governmental organizations. These organizations frequently make significant financial, human, and time efforts to validate measuring methods for use in public policy.	Scales that are seldom used or were constructed a long time ago should not be used carelessly. This might indicate inadequate psychometric abilities.	If various factorial structures (higher order or bifactor models) are anticipated for the same scale, justify the comparisons to avoid using CFA as an exploratory technique (Jackson et al., 2009).
	Exhaust all research efforts to find the best measuring model. Schumacker et al. (2021) present a wide list of tests and measures as a starting point, in addition to specialist periodicals, books, technical reports, and dissertations or theses. There are various systematic reviews (or meta-analyses) of the literature on measurement models for a particular topic, area, or concept. You could come upon one that provides greater credence to your research. Always go over the original scale articles and any later applications. Kline (2016, p.29) presents an interesting checklist for evaluating potential measures.	It is not recommended to quickly translate and retranslate a scale that has been used in another language. It is not recommended to apply SEM based on variances to a single sample that also analyzes the structural model, as is usually done.	Determine your favorite model and justify your choice by outlining the merits of your selected model over equal and competing theoretical theories (Jackson et al., 2009).
	Consider incorporating control questions (e.g., choosing "strongly agree" on this items) and tracking response time to the questionnaire to assess respondents' engagement (Collier, 2020).	Do not withhold information on alternate factorial structures for the same measurement model (bifactor or higher order models). If these alternative structures have any theoretical justification and relevance, they may be able to save the researcher's research, even if they believe it will necessitate more work (Crede & Harms, 2019).	Give a graphical representation of the models that were tested (Jackson et al., 2009; Kline, 2016). A priori define and describe suspected cross-loadings based on existing empirical findings (Nye, 2022).

Table 3 (continued)

Step	Best practices	What not to do	What to report
(2) Pre-processing	Remove responses from respondents who have effectively abandoned the survey. Delete observations from respondents who chose the control questions incorrectly and finished the questionnaire in considerably less time than expected (Collier, 2020).	Avoid single-imputation (listwise, pairwise, mean-substitution, regression substitution and pattern matching) (Brown, 2015). All single-imputation methods tend to underestimate error variance, especially if the proportion of missing observations is relatively high (Kline, 2016). Some of these techniques may cause identification problems. A few missing values, such as < 5% in the total data set, may be of little concern (Kline, 2016).	Maintain transparency about missing items. Every researcher experiences this problem, and when it occurs in tiny numbers, it is manageable and does not discredit the study (Crede & Harms, 2019; Kline, 2016).
	Address missing data before beginning the CFA by multiple imputation (Brown, 2015). The alternative suggested treatment for missing values (Full Information ML - FIML), which is included in many software packages, including JASP, is for continuous data. The handling of missing values for ordinal items in CFA is still evolving.	In CFAs, which generally feature answer options ranging from 1 to 5, univariate outliers are infrequent. Is it unusual to answer 1 or 5? It makes more sense to investigate anomalies and engagement in the following ways: i) identifying individuals who answered all 1 or all 5 when other answer choices were expected; or (ii) identifying persons who demonstrate response patterns, such as answering all 1 at the beginning and all 5 at the end.	Implement and report on ways for determining respondent engagement.
	Calculate the variability (standard deviation) for each participant to see if they responded the same way to all items. A standard deviation of zero or near zero may suggest a lack of engagement. Cases with little variability do not contribute to the estimations, therefore evaluate each instance separately and consider exclusions (Collier, 2020).	Items should not be parcelized because: i) there are several ways to do so; and ii) doing so may hide multidimensionality (Crede & Harms, 2019).	Present evidence of multicollinearity prior to beginning CFA (Whittaker & Schumacker, 2022)
	Calculate the variance inflation factor (VIF) for each item and consider removing those with a high degree of multicollinearity (VIF > 10) (Kline, 2016, Whittaker & Schumacker, 2022).		Declare which packages and/or programs were utilized, as well as the version number (Flake & Fried, 2020; Jackson et al., 2009).
	Only consider multivariate outliers. Identify them and then evaluate their exclusion based on the profile of your sample (Kline, 2016, Whittaker & Schumacker, 2022). The correlation residuals (the gap between estimated and observed correlation) are the most interesting outliers since they apply to items rather than individuals.		Make raw data or the items' covariance/correlation matrix public (open data) (Crede & Harms, 2019; Flake et al., 2022; Flake & Fried, 2020).
(3) Estimation process	Employ estimate techniques that take into account the data's ordinal measurement level: the OLS family (Ordinary Least Squared), specifically ULS (Unweighted LS) or DWLS (Diagonally Weighted LS), and, lastly, the WLS (Weighted LS) (Nye, 2022). According to simulation studies, ULS produces the greatest outcomes in small samples (Forero et al. 2009; Holgado-Tello et al., 2016, 2018; Kiliç & Dogan, 2021; Kiliç et al. 2020; Li, 2016a, b; Morata-Ramírez & Holgado-Tello, 2013; Savalei & Rhemtulla, 2013; Yang-Wallentin et al., 2010).	Even in its robust variant (MLR), the maximum likelihood (ML) technique should only be used for continuous data (or data that may be understood as such) with minimal deviations from multivariate normality (Bandalos, 2014; Beauducel & Herzberg, 2006; DiStefano & Morgan, 2014; Flora & Curran, 2004; Forero et al., 2009; Holgado-Tello et al., 2016, 2018; Kiliç et al., 2020; Kiliç & Dogan, 2021; Li, 2016a, b; Mîndrilă, 2010; Morata-Ramírez & Holgado-Tello, 2013; Nye, 2022; Savalei & Rhemtulla, 2013; Yang-Wallentin et al., 2010)	Document and justify the estimating approach used, taking into consideration the available data and the model to be adjusted (Crede & Harms, 2019). Declare all difficulties observed during the estimating process, such as algorithm non-convergence and unidentified models (Nye, 2022). To determine the scale of the latent variable, specify whether you fixed the variance or used a marker variable (=1) (Jackson et al., 2009).
	The DWLS approach, also known as WLSMV (WLS Mean and Variance), is suggested for multivariate normality violations and can employ the same correction measures as a CFA (Nye, 2022). In general, the WLS approach requires a bigger sample size (Beauducel & Herzberg, 2006; Jackson et al., 2009; Li, 2016b; Nalbantoğlu-Yılmaz, 2019, Savalei & Rhemtulla, 2013). Robust versions of the estimators, such as Robust DWLS or RDWLS, should be used (DiStefano & Morgan, 2014, Flora & Curran, 2004; Holgado-Tello et al., 2016, 2018; Kiliç et al., 2020; Lei, 2009; Li, 2016a)		Report parameters that are both standardized and non-standardized (Nye, 2022).

Table 3 (continued)

Step	Best practices	What not to do	What to report
(4) Model fitting	Look for standardized factor loadings greater than one or negative variances in Heywood scenarios (Nye, 2022).	Do not use χ^2 unless it has been properly adjusted (for example, mean and variance corrected) (Crede & Harms, 2019).	Report the cutoffs utilized (Jackson et al., 2009). If from the DFI technique, which is only accessible in the R environment for categorical data and/or Likert-type items (Wolf & McNeish, 2023), whose package (dynamic) and calculation functions (catHB and catOne) employ lavaan objects (McNeish, 2023a).
	Every global index has restrictions. Analyze the default global indices [SRMR, RMSEA, and CFI] together rather than individually. If any global index reveals a misfit but only minor deviations from the cutoffs, more study may be required to establish that the misfit is attributable to data characteristics or an index restriction rather than a model misspecification (Nye, 2022). In this respect, conventional global fit indices should be seen as an effect size, as originally intended (McNeish & Wolf, 2023), rather than an ad hoc hypothesis test in which a choice to reject or not reject the model is made (McNeish, 2023b). Give higher importance to SRMR, which looks to be more robust in respect to the estimator used (Shi, Maydeu-Olivares, et al., 2020; Shi & Maydeu-Olivares, 2020).	Surpassing a set of cutoff values should not be the main rationale for accepting a model (Xia & Yang, 2019). Do not use less conservative traditional cutoffs [RMSEA ≤.08, CFI ≥.90 and SRMR ≤.08]. Use more conservative traditional cutoffs [SRMR ≤.06, RMSEA ≤.06 and CFI ≥.95] carefully, because recent studies show that DWLS and ULS result in lower RMSEA values (≤.06) and higher CFI values (≥.95) than ML (Groskurth et al., 2023; McNeish, 2023b; Xia & Yang, 2019), the estimator used to define traditional cutoffs. Furthermore, previous cutoff investigations were based on simulation experiments that took into consideration a maximum structure with three factors, fifteen items, and factor loadings ranging from 0.7 to 0.8. (McNeish & Wolf, 2023). Furthermore, traditional cutoff approaches are affected by the number of items and factors, model degrees of freedom, magnitude of factor loadings, misfit type, and missing data (Groskurth et al., 2023; McNeish, 2023b).	Report the scaled adjusted χ^2 and the degrees of freedom (*df*) (Crede & Harms, 2019).
	The Dynamic Fit Index (DFI) appears to be a good method for evaluating the global adjustment index, but its application must be careful, since its recentness requires more positive empirical data (McNeish, 2023a). It is a simulation-based recommendation with specified cutoff criteria for the model under consideration. The reasoning of this technique is that it mirrors the strategy used to calculate the cutoffs in traditional models, but adjusts the simulation design to the model subspace that corresponds to the empirical model under review (McNeish, 2023b; McNeish & Wolf, 2022, 2023; Wolf & McNeish, 2023).		Report at least the RMSEA, CFI, and SRMR global fit indices. Because you will be using scaled χ^2, these measurements must also be modified to account for the technique employed (Nye, 2022).
			Report at least McDonald's Omega, but ideally GLB or another composite reliability measure as well (Trizano-Hermosilla & Alvarado, 2016).
		Do not overlook huge variations in model fit: only minor deviations from the cutoffs may be justified; a significant misfit is more likely to suggest a fundamental problem with the model than a restriction of the fit indices (Nye, 2022).	
	Local adjustment indicators such as low factor loading, cross-loadings, residual correlation (>0.10), modification indices (>3.84), and estimated parameter change (EPC) should be evaluated (Nye, 2022).	Cronbach's Alpha should not be used as a measure of reliability (Bell et al., 2023; Cho, 2022; Dunn et al., 2014; Flora, 2020; Goodboy & Martin, 2020; Green & Yang, 2015; Hayes & Coutts, 2020; Kalkbrenner, 2023; McNeish, 2018; Trizano-Hermosilla & Alvarado, 2016).	Report all model adjustment decisions in accordance with the evaluation of local fit measures (Flake & Fried, 2020). (low factor loading, cross-loadings, residual correlation, modification indices and EPC). Indicate the magnitude of residual correlations and modification indices (Crede & Harms, 2019).
	In the context of the adjusted CFA model, use McDonald's Omega or GLB (Greatest lower limit) as the reliability coefficient (Bell et al., 2023; Cho, 2022; Dunn et al., 2014; Flora, 2020; Goodboy & Martin, 2020; Green & Yang, 2015; Hayes & Coutts, 2020; Kalkbrenner, 2023; McNeish, 2018; Trizano-Hermosilla & Alvarado, 2016).		

Table 3 (continued)

Step	Best practices	What not to do	What to report
(5) Model comparisons and modifications	If the model's fit measurements are not satisfactory, it may be required to investigate alternatives or adjustments. It is critical, however, to guarantee that these changes are conceptually justifiable. For example, one may look into if comparable problems have been observed in past study (Nye, 2022).	When comparing models, it is critical not to utilize χ^2 without sufficient adjustment.	Indeed, all modifications made to the measurement model, no matter how questionable, must be notified (Flake & Fried, 2020). To prevent being questioned about using CFA for exploratory purposes, these possible changes should ideally be understood in before (Flake & Fried, 2020). However, such changes occur up to 50% of the time (Flake et al., 2017; Nye, 2022).
	The adjusted χ^2 (scaled) should only be used for nested model comparisons.	Believing that no changes should be made to a measuring model that has been validated in earlier research and continuing with research that has significant misfit can lead to issues. Misfit in the CFA can spread to the structural model, and to some extent, some CFA modifications can fix difficulties in the estimate of structural connections, which are normally of most importance in a typical CFA (Flake et al., 2017; Jackson et al., 2009).	Exclusions of items are easily identifiable from survey results. Addressing relationships between things, on the other hand, is more difficult. As a result, it is critical to disclose any and all inter-item correlations projected to improve model fit, as well as offer the theoretical explanation for these correlations (Nye, 2022).
	Regardless of the presence of accumulated validity evidence in favor of the measurement model under study, if substantial doubts about the theoretical factorial structure and/or level of misfit persist, the researcher should proceed to compare competing models using simulations within the Power Analysis approach (Leite et al., 2023; Feng & Hancock, 2023; Kyriazos, 2018; Zhang & Yuan, 2018).	Drastic changes in the measuring model (more than 20%) should be avoided (Kline, 2016). Consider returning to prior stages, such as utilizing an EFA or choosing a different scale. Consider proceeding by trying to adjust Exploratory SEM (ESEM) models to efficiently model cross-loadings (Crede & Harms, 2019). Remember that you may potentially alter the model until it gets a perfect fit (saturated model), but this would compromise external validity.	Transparency is essential when reporting information regarding the measurement model's fit (Flake et al., 2022; Flake & Fried, 2020). The argument that journal space is restricted is no longer true, because researchers may now make any sort of material permanently available in one of the multiple open access repositories and mention it in the study article.
		Correlations between indicator mistakes are only recommended if: i) the items are part of the same concept; ii) there is theoretical reason; iii) the items are antonyms; and iv) reverse items exist. In general, this technique should be justified primarily to address the Common Method Variance (CMB) issue (Nye, 2022).	

Table 3 (continued)

Step	Best practices	What not to do	What to report
(6) Power Analysis	Sample planning should be done before to data collection (a priori), taking into account the global conceptual model (problem) of the investigation. From this standpoint, this should be one of the initial steps (Kyriazos, 2018; Nye, 2022).	It is not appropriate to justify the sample size decision with a "rule of thumb," as there is a rule for every scenario, regardless of the researcher's effort. There are general guidelines to suit all tastes: i) studies with a minimum size (N) of 50, 100, 200, 300, 400, 500, and 1000; or ii) requirements for number of observations/number of parameters (N/p) of 20/1, 10/1, and 5/1. Ordinal data, as estimated by DWLS or WLS, need a higher sample size. Larger samples are required for more complicated models with low psychometric quality. In the context of SEM, general criteria for sample size needs are impracticable (Kyriazos, 2018).	Indeed, it is suggested in a typical CFA to show the following simulation statistics during power analysis: i) estimated parameter bias; ii) standard error bias; iii) coverage; and iv) power. The first two requirements should not exceed 10% in absolute terms for all parameters or 5% for key parameters (factor loadings + factor correlations). Coverage should be between 0.91 and 0.98, with power over 0.80 (Kyriazos, 2018; Muthén & Muthén, 2002).
	However, in typical CFAs, the measuring model is the "means" rather than the "goal" of the investigation (Flake et al., 2022). To conduct a more thorough sample calculation effort, the researcher must take into account both the measurement models and the structural model. For a typical CFA, at least an a posteriori power analysis is recommended in this context (Beaujean, 2014). A more effective and current technique is to tailor the sample power for each study, taking into account the research's specific context and focal questions (Feng & Hancock, 2023). Power Analysis in SEM calculations are difficult and rely on several factors (Jak et al., 2021). In particular, in a typical CFA when data is ordinal and ML is not advised, analytical answers are problematic (Feng & Hancock, 2023), and should only be utilized as a starting point. Only Monte Carlo Simulation can provide a more accurate assessment of sampling power in a typical CFA, regardless of whether the sampling power analysis is a priori or post hoc (Feng & Hancock, 2023; Leite et al., 2023).	Analytical solutions such as the Satorra-Sarris approach and/or the RMSEA misfit method should not be employed. Although these solutions are elegant, they are based on asymptotic features of the test statistic and certain distributions underlying the discrepancy function (e.g., multivariate normality) (Feng & Hancock, 2023; Zhang & Yuan, 2018).	The practice of publishing the power analysis for the measurement model(s) is thought to help in science's replicability and reproducibility (Flake et al., 2017, 2022), adding to the accumulation of evidence of the internal structure's validity (Flake & Fried, 2020). A low power identified in the context of the CFA under examination may act as a caution indication for inferences derived from the complete model.

A typical CFA in applied social sciences is defined as research that tries to adapt a common factor model. The measurement scale in this paradigm has a specified factorial structure with established psychometric features. These studies' data are typically ordinal and/or lack a multivariate normal distribution. These situations are common in studies in which the reflective measurement model is used as a "means" rather than a "end." Such studies frequently examine questionnaires that contain multiple Likert-type items, each with at least five response alternatives (Flake et al., 2017).

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